

Project Design Phase
Solution Architecture

Date	29 September 2026
Team ID	SWTID-2026-9956
Project Name	Auto Ticket Classification Using Flow Designer
Maximum Marks	4 Marks

Solution Architecture:

1. Find the Best Technology Solution to Solve Existing Business Problems

The proposed solution uses ServiceNow Flow Designer to automate the classification and notification of School IT support tickets. The system automatically reads the ticket Short Description when a new record is created and identifies the appropriate Category and Subcategory based on predefined conditions.

The selected technology solution consists of:

- ServiceNow Platform – Central platform for managing IT support tickets.
- ServiceNow Ticket Table – Stores Caller, Short Description, Description, Category, Subcategory, State, Assigned Group, and Assigned To.
- Flow Designer – Automates ticket classification and processing.
- Conditional Logic – Identifies the issue type from the Short Description.
- Update Record Action – Automatically assigns Category and Subcategory.
- Send Email Action – Sends an automated confirmation to the Caller.

The classification rules are:

Wi-Fi / Network → Network → Wi-Fi

Projector → Hardware → Projector

Forgot Password → Access → Forgot Password

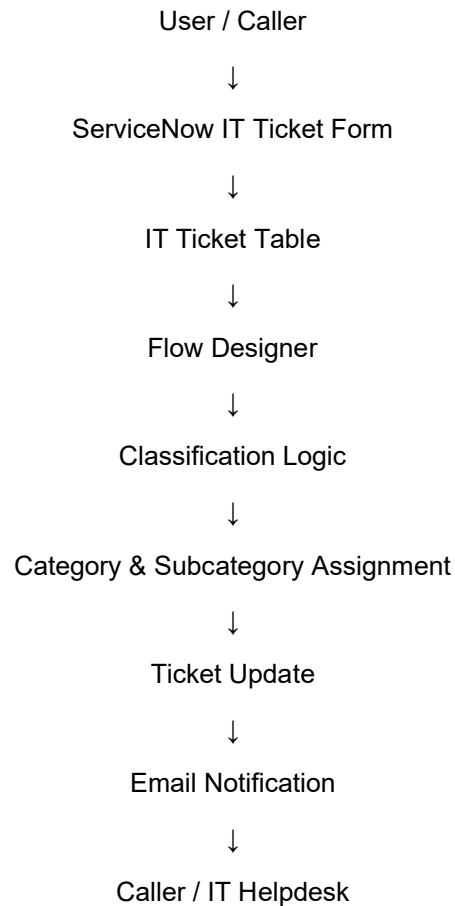
Slow Computer → Performance → Slow Computer

This solution addresses the existing problem of manual ticket classification and reduces repetitive work for IT Helpdesk staff. The provided implementation uses a Record Created trigger with the condition Category is Empty.

2. Structure, Characteristics and Behaviour of the Software:

Software Structure

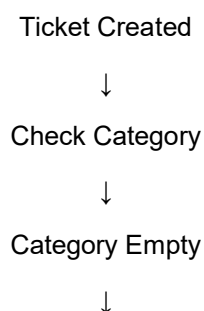
The solution consists of the following major components:



Software Characteristics

- **Automated:** Ticket classification is performed automatically.
- **Structured:** Ticket information is maintained using predefined fields and choices.
- **Consistent:** Standard Category and Subcategory values are used.
- **Maintainable:** Classification conditions can be modified or extended through Flow Designer.
- **Scalable:** Additional issue types and automation rules can be added.
- **User-friendly:** Users only need to provide the issue description; classification is handled by the system.
- **Integrated:** Classification, ticket updating, and email notification are handled within the ServiceNow platform.

Software Behaviour:



Read Short Description



Identify Issue



Assign Category



Assign Subcategory



Update Ticket



Send Email Notification



Helpdesk Processes Ticket

The implementation uses conditional branches for Wi-Fi, Projector, Forgot Password, and Slow Computer issues.

3. Features, Development Phases and Solution Requirements

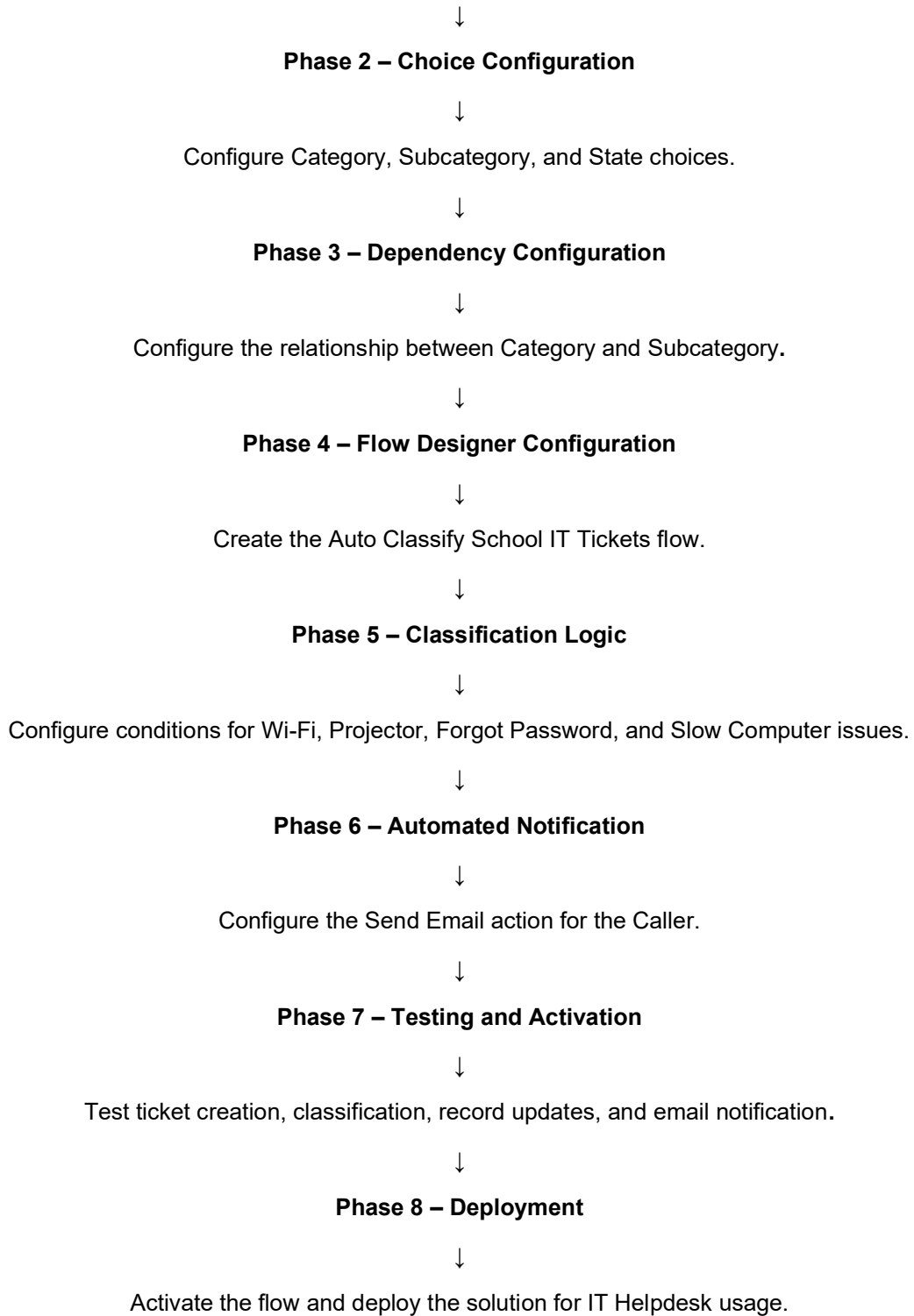
Feature	Description
IT Ticket Creation	Allows users to create IT support tickets.
Automatic Classification	Identifies the issue from the Short Description.
Automatic Category Assignment	Assigns the appropriate Category without manual intervention.
Automatic Subcategory Assignment	Assigns the appropriate Subcategory based on the identified issue.
Ticket Update	Automatically updates the ticket record with classification details.
Email Notification	Sends confirmation to the Caller after ticket processing.
Standardized Data	Maintains consistent ticket information.
Future Extensibility	Allows additional classification rules and issue types to be added.

Development Phases

Phase 1 – Ticket Table Configuration



Create and configure required ticket fields such as Caller, Category, Subcategory, Short Description, Description, State, Assigned Group, and Assigned To.



Solution Requirements

Requirement	Description
Automatic Classification	System must classify tickets based on issue description.
Category Assignment	System must automatically assign the appropriate Category.
Subcategory Assignment	System must automatically assign the appropriate Subcategory.

Dependent Choices	Subcategory options must correspond to the selected Category.
Structured Storage	Ticket information must be stored in predefined fields.
Automated Notification	System must send an email notification to the Caller.
Maintainability	Classification rules should be easy to modify.
Scalability	System should support additional categories, subcategories, and classification rules.

4. Specifications for Defining, Managing and Delivering the Solution:

Technical Specifications

Specification	Implementation
Platform	ServiceNow
Automation Tool	Flow Designer
Trigger	Record Created
Trigger Condition	Category is Empty
Input	Short Description
Classification Method	Conditional Flow Logic
Data Update	Update Record
Notification	Send Email
Email Recipient	Caller
Classification Output	Category and Subcategory
Ticket Storage	ServiceNow IT Ticket Table
Administration	ServiceNow Configuration and Flow Designer

SOLUTION ARCHITECTURE DIAGRAM

